CIRCUITCYCLE ACADEMIC REPOSITORY // EWM-TRK-01

HIMACHAL PRADESH

E-Waste Management: Performance & Policy Assessment

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With United Nations Sustainable Development Goals (SDG) Alignment

Regulatory Framework: E-Waste (Management) Rules, 2022 (MoEFCC / CPCB) & HPSPCB

Global Framework: UN Sustainable Development Goals – Agenda 2030

Evaluation Model: State E-Waste Management 100-Point Scorecard

Date of Assessment: July 2026

STATE SCORE: 67.3 / 100 — “GOOD” (Band: 65–84)

Live Interactive Dashboard:

himachalpradesh-sdg-tracker.lovable.app

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Executive Summary

Himachal Pradesh achieves a State Score of 67.3 out of 100 on the State E-Waste Management 100-Point Scorecard, placing it in the “GOOD” performance band (65–84). The state's principal strength is its formal recycling infrastructure, concentrated in the Baddi-Barotiwala-Nalagarh (BBN) industrial corridor of Solan district, where five HPSPCB-authorised dismantling/recycling facilities offer a combined installed capacity of 7,384 tonnes/year — more than double the state's estimated annual e-waste generation of ~3,566 tonnes/year.

The scorecard also surfaces two clear weak points that pull the overall score down from the “Excellent” band: actual collection performance (46.6% of generated e-waste is collected against a 100% benchmark) and informal-sector formalisation (none of the four identified informal scrap clusters are yet fully formalised, though partial skilling/registration is underway at the largest one). Awareness and outreach activity, while reaching an estimated 159,050 people through six logged programmes, falls short of the 20-programmes/year benchmark used in this assessment.

This report presents the full district-wise generation inventory, collection log, authorised-recycler register, EPR compliance tracker, informal-hotspot register, environmental/health monitoring log and awareness-programme log that feed the scorecard, together with the accompanying UN SDG alignment matrix, key gaps, and policy recommendations. All figures are also available as a live, filterable dashboard — see the link on the cover page and in Section 3.

1. Methodology & Data Sources

This assessment follows a standard “State E-Waste Management Tracker” template (seven data sheets plus an auto-calculating scorecard) and applies it to Himachal Pradesh's twelve districts. In the interest of transparency, data used in this report falls into two categories:

Verified / publicly documented: the state's 12-district administrative structure and approximate populations; the applicable regulatory framework (E-Waste (Management) Rules, 2022 under MoEFCC/CPCB, administered in-state by the Himachal

Pradesh State Pollution Control Board, HPSPCB); the existence and general location of the Baddi-Barotiwala-Nalagarh (BBN) industrial belt in Solan district as the state's principal electronics/light-manufacturing and formal e-waste recycling hub; and HPSPCB's own submissions (e.g. in NGT-related compliance filings) that authorised e-waste dismantling/recycling capacity in the state is concentrated among a small number of Solan-district facilities, including Shivalik Solid Waste Management Ltd. and Ortech India Corporation.

Estimated / illustrative (for this assignment and dashboard prototype): district-wise generation splits by equipment category, collection-log quantities, specific EPR compliance case records, informal-hotspot names and worker counts, environmental-monitoring readings, and awareness-programme attendance. HPSPCB and CPCB do not currently publish a single, granular, district-level open dataset covering all of these fields for Himachal Pradesh, so these figures were modelled using population-weighted, proportionate estimates anchored to national CPCB e-waste generation benchmarks and plausible, realistic ranges for a small hill state. They should be treated as a structured planning estimate — a credible starting model for the tracker — rather than as verbatim government-published statistics, and should be refreshed with primary HPSPCB/CPCB/EPR-portal data (or RTI-sourced figures) as and when that becomes available.

All totals, percentages and the final scorecard in this report are computed directly and consistently from the underlying tracker workbook (formulas, not manual arithmetic), so every number below is internally consistent and traceable to a single source of truth: the accompanying Excel workbook “Himachal Pradesh E-Waste Tracker.xlsx”.

2. Live Interactive Dashboard

The data in this report has been built out into a live, interactive web dashboard so the scorecard, district maps and category drill-downs can be explored directly rather than only read as a static document:

→ https://himachalpradesh-sdg-tracker.lovable.app/

Note: the dashboard mirrors the figures in this report and the accompanying workbook. If the two are ever updated at different times, treat the most recently edited source as authoritative.

3. State Profile & District E-Waste Generation Dynamics

3.1 Demographic & Economic Overview

Himachal Pradesh has an estimated population of ~7.17 million spread across 12 districts. Economic activity relevant to e-waste generation and recycling capacity clusters into three broad zones:

Industrial manufacturing belt: Baddi, Barotiwala and Nalagarh (Solan district) and Paonta Sahib (Sirmaur district) — home to pharmaceutical, electronic-component and light-manufacturing units, and to essentially all of the state's authorised e-waste recycling capacity.

Urban & tourism hubs: Shimla, Dharamshala (Kangra), Mandi, and Kullu-Manali — the largest sources of consumer/household e-waste and the main sites of informal kabadiwala (scrap-dealer) activity.

High-altitude rural districts: Bilaspur, Hamirpur, Una, Chamba, Kinnaur and Lahaul & Spiti — low generation volumes but disproportionately high per-tonne collection costs due to terrain and seasonal road closures.

3.2 District-Wise E-Waste Generation Inventory (kg/year)

District

Population

IT & Telecom

Consumer Elec.

Large/Small Equip.

Lighting	
Batteries	
Other	
Total (kg)	
Solan	
6,10,000	
7,20,000	
3,80,000	
2,50,000	
65,000	
95,000	
40,000	
15,50,000	
Shimla	
8,60,000	
2,80,000	
1,60,000	
1,00,000	
25,000	
35,000	
15,000	
6,15,000	
Kangra	
15,80,000	
2,40,000	
1,40,000	
85,000	
20,000	
30,000	
12,000	
5,27,000	
Mandi	
10,40,000	
1,60,000	
95,000	

60,000

15,000

22,000

8,000

3,60,000

Sirmaur

5,60,000

1,20,000

70,000

45,000

10,000

15,000

5,000

2,65,000

Una

5,50,000

45,000

30,000

20,000

5,000

8,000

3,000

1,11,000

Hamirpur

4,70,000

22,000

15,000

10,000

2,500

4,000

1,500

55,000

Kullu

4,60,000

18,000

12,000

8,000

2,000

3,000

1,000

44,000

Bilaspur

3,90,000

8,000

5,500

3,500

1,000

1,500

500

20,000

Chamba

5,30,000

5,000

3,500

2,000

500

800

200

12,000

Kinnaur

85,000

2,000

1,200

800

200

300

100

4,600

Lahaul & Spiti

32,000

1,000

600

400

100

150

50

2,300

STATE TOTAL

71,67,000

16,21,000

9,12,800

5,84,700

1,46,300

2,14,750

86,350

35,65,900

Total generation ≈ 3,565.9 MT/year; Solan district alone accounts for ~43.5% of the state total, reflecting concentrated industrial/IT-equipment turnover in the BBN belt.

4. State E-Waste Management Scorecard

The scorecard weights seven performance categories to 100 points. Achievement is calculated as (Achieved Value ÷ Benchmark), capped at 100%, then multiplied by the category weight.

#

Category

Wt.

Benchmark

Achieved

Achv. %

Score

1

Generation & Inventory

15

5 districts w/ data

12 / 12 districts

100.0%

15.0

2

Collection Performance

20

100% of generation

46.6% collected

46.6%

9.3

3

Formal Recycling & Treatment

20

Capacity ≥ generation

207.1% (7,384 vs 3,566 MT)

100.0%

20.0

4

Regulatory Compliance

15

100% compliant

4 / 6 producers (66.7%)

66.7%

10.0

5

Informal Sector Integration

10

100% formalized

0 / 4 fully formalized

0.0%

0.0

6

Environmental & Health Safeguards

10

100% sites monitored

4 / 4 hotspots monitored

100.0%

10.0

7

Awareness & Outreach

10

20 programs/yr

6 programs logged

30.0%

3.0

TOTAL SCORE

100

—

—

—

67.3

Score bands used: 85–100 Excellent · 65–84 Good · 45–64 Needs Improvement · 0–44 Critical. Himachal Pradesh: 67.3 → GOOD.

4.1 UN SDG Alignment & Impact Matrix

UN SDG Target

Domain

HP Performance

Rating

SDG 12.4

Category-wise inventory completeness

5 CPCB waste categories tracked across all 12 districts

HIGH (100%)

SDG 9.4 & 12.5

Installed recycling capacity vs. recovery

7,384 MT/yr capacity vs. 3,566 MT/yr generation; 88.4% avg. material recovery

EXCELLENT (100%)

SDG 11.6

District EPR / municipal collection rate

1,662.3 MT collected of 3,565.9 MT generated (46.6%)

MODERATE (46.6%)

SDG 8.8 & 1.3

Informal-sector formalisation & safety

70 of 265 informal workers registered/trained; 0 of 4 hotspots fully formalised

NEEDS IMPROVEMENT (0–26%)

SDG 3.9 & 6.3

Soil/water toxin monitoring & worker health

4 of 4 identified hotspots monitored; 410 workers/residents screened

GOOD (100%)

SDG 13.3 & 12.8

Climate & circularity awareness

6 programmes logged, reaching ~159,050 residents

NEEDS IMPROVEMENT (30%)

SDG 16.6

Governance transparency & enforcement

6 of 6 planned inspections conducted; 4 of 6 EPR producers compliant (66.7%)

GOOD (66.7–100%)

5. District Collection Performance

District

Generated (kg)

Collected (kg)

Centers

EPR Target (kg)

Collection Rate

Solan

15,50,000

8,20,000

14

10,00,000

52.9%

Shimla

6,15,000

3,20,000

6

4,00,000

52.0%

Kangra

5,27,000

2,10,000

5

3,50,000

39.8%

Mandi

3,60,000

1,40,000

4

2,40,000

38.9%

Sirmaur

2,65,000

1,10,000

3

1,80,000

41.5%

Una

1,11,000

35,000

2

70,000

31.5%

Hamirpur

55,000

12,000

1

35,000

21.8%

Kullu

44,000

8,500

1

25,000

19.3%

Bilaspur

20,000

4,000

1

12,000

20.0%

Chamba

12,000

2,000

1

8,000

16.7%

Kinnaur

4,600

600

0

3,000

13.0%

Lahaul & Spiti

2,300

200

0

1,500

8.7%

STATE TOTAL

35,65,900

16,62,300

38

23,24,500

46.6%

State collection rate = Collected ÷ Generated = 1,662,300 ÷ 3,565,900 = 46.6%. Collection rate falls sharply outside the Solan/Shimla/Kangra urban-industrial core, confirming the “mountain logistics” gap discussed in Section 8.

6. Authorised Recycling & Dismantling Infrastructure

Himachal Pradesh's authorised e-waste processing capacity is concentrated almost entirely in the Baddi-Barotiwala-Nalagarh (BBN) industrial belt of Solan district:

Recycler / Dismantler

Authorisation No.

District

Capacity (t/yr)

Processed (t)

Recovery %

Valid Till

M/s Shivalik Solid Waste Mgmt. Ltd. (Unit-I)

HPSPCB/EWM/NLG/01

Solan (Nalagarh)

5,200

2,450

92.5%

31-03-2028

M/s Shivalik Solid Waste Mgmt. Ltd. (Unit-II)

HPSPCB/EWM/NLG/02

Solan (Nalagarh)

1,000

480

91.0%

31-12-2027

M/s Eco Works Management Pvt. Ltd.

HPSPCB/EWM/NLG/03

Solan (Nalagarh)

540

290

88.5%

31-08-2027

M/s Ortech India Corporation

HPSPCB/EWM/BDI/04

Solan (Baddi)

500

210

86.0%

30-06-2027

M/s Sahni Enterprise

HPSPCB/EWM/PRW/05

Solan (Parwanoo)

144

85

84.0%

31-10-2026

STATE TOTAL

—

—

7,384

3,515

88.4%

—

Installed capacity (7,384 t/yr) exceeds estimated annual generation (~3,566 t/yr) by roughly 107%, i.e. the state's formal processing capacity is more than double its own generation — though a material share of throughput likely originates from neighbouring industrial clusters given Solan's position on the Punjab/Chandigarh industrial corridor. Facility names, districts and authorisation structure are consistent with HPSPCB's own public filings; capacity, throughput and recovery-rate figures are illustrative planning estimates pending verification against HPSPCB's current consent-to-operate register.

7. EPR Producer Compliance Tracker

Producer / EPR Entity

EPR Reg. No.

Status

Inspected

Penalty

HP Industrial Tech Solutions

EPR/2024/HP/001

Compliant

Yes

No

Baddi Electricals Pvt. Ltd.

EPR/2024/HP/002

Compliant

Yes

No

Himalayan Consumer Durables

EPR/2024/HP/003

Non-Compliant

Yes

Yes (Resolved)

Shimla Telecom Services

EPR/2024/HP/004

Compliant

Yes

No

Solan Green Electronics

EPR/2024/HP/005

Compliant

Yes

No

Paonta Power Systems

EPR/2024/HP/006

Non-Compliant

Yes

Yes (Pending)

Summary: 6 producers registered · 4 compliant (66.7%) · 6/6 planned inspections conducted · 2 penalties/show-cause notices issued · 1 resolved, 1 pending.

8. Informal Sector Hotspots

Hotspot / Cluster

District

Status

Est. Workers

Registered/Trained

Sabzi Mandi Old Scrap Market

Solan

Partially Formalized

120

45

Ganj Bazaar Kabadi Cluster

Shimla

Unformalized

65

12

Kachhari Adda Scrap Hub

Kangra

Unformalized

45

8

Mandi Bus Stand Kabadi Mandi

Mandi

Unformalized

35

5

TOTAL

—

0 fully formalised

265

70 (26.4%)

Sabzi Mandi Old Scrap Market (Solan) is the largest cluster and the only one with an active formalisation programme underway (skill training + HPSPCB ID-card registration); it is treated as “Partially Formalized” rather than “Formalized”, so it does not yet score against the scorecard's formalisation benchmark.

9. Environmental & Health Monitoring

Site

District

Monitoring Type

Date

Result

Screened

Baddi-Nalagarh Industrial Drain

Solan

Soil & surface water (Pb, Cd, Hg)

15-04-2026

Within CPCB limits

180

Solan Scrap Cluster Surroundings

Solan

Soil heavy-metal testing

22-05-2026

Trace Pb, below threshold

95

Shimla Ridge Solid Waste Site

Shimla

Air & runoff water quality

10-06-2026

Compliant (BIS)

60

Paonta Sahib Industrial Area

Sirmaur

Soil & groundwater

28-06-2026

Normal pH/metals

75

TOTAL — 4 sites, 4 health camps

410

All four monitored sites returned results within or close to CPCB/BIS permissible limits at the time of testing; the Solan scrap-cluster site (trace lead) is flagged for a 6-month follow-up test.

10. Awareness & Outreach Programmes

Programme

Clean Hills E-Waste Drive	Shimla	15-03-2026	RWA/Community Drive	3,500
Green School E-Waste Awareness	Kangra (Dharamshala)	12-04-2026	School Collection Drive	2,200
BBN Industrial Circularity Workshop	Solan	28-04-2026	Corporate/EPR Seminar	450
Mandi College E-Waste Hunt	Mandi	14-05-2026	Youth Recycling Campaign	1,800
Paonta Industrial Take-Back Expo	Sirmaur	05-06-2026	E-Waste Collection Camp	1,100
State E-Waste Awareness Month	Statewide (12 Districts)	01-06-2026	Digital & Print Campaign	1,50,000
TOTAL — 6 programmes				

159050

The single State E-Waste Awareness Month digital/print campaign accounts for ~94% of total reported reach; in-person, district-level engagement (excluding this campaign) totals only ~9,050 people — the underlying gap behind the Awareness & Outreach category's low score.

11. Key Gaps & Topographical Challenges

Mountainous terrain & logistics barriers: High per-tonne freight costs in remote districts (Kinnaur, Lahaul & Spiti, Chamba) suppress collection rates to 13–17% even though generation is low in absolute terms — directly impacting SDG 11.6.

Informal-sector exposure (SDG 8.8): Unorganised scrap handlers (kabadiwalas) in Shimla and Kangra operate without registration or PPE; only the Solan cluster has a formalisation programme underway, and even there it is partial.

Rural awareness deficit: Industrial units in the BBN belt file EPR returns reasonably well, but rural households have no drop-off infrastructure or incentive-based take-back schemes (SDG 12.8); most reported “reach” is passive media exposure, not verified take-back participation.

Collection-target design gap: The state's own EPR collection targets (2,324,500 kg) sit well above what is actually being achieved (1,662,300 kg), meaning even full delivery against current targets would only reach ~65% of generation — targets themselves may need revising upward over time.

12. Strategic Policy Recommendations

Mountain e-waste drop-off kiosks (SDG 11.6): Deploy SPCB-assisted collection kiosks at district headquarters in Kullu, Chamba, Kinnaur and Lahaul & Spiti, timed around the annual road-access window, to intercept high-altitude waste before it reaches informal channels.

Formalise informal workers via Self-Help Groups (SDG 8.8 & 1.3): Extend the Sabzi Mandi (Solan) model — SPCB identity cards, PPE kits and per-kg routing commissions to authorised Nalagarh recyclers — to the Ganj Bazaar (Shimla), Kachhari Adda (Kangra) and Mandi Bus Stand clusters.

Mandatory corporate circularity audits (SDG 12.6): Introduce quarterly (not just annual) verification of Form-2 EPR returns filed by BBN-belt manufacturers, closing the gap that currently leaves 2 of 6 tracked producers non-compliant.

“Dispose & Win” tourist-spot drives (SDG 12.8 & 13.3): Introduce incentive-based collection bins in high-footfall tourist areas (Shimla Ridge, Manali Mall Road, Dharamshala) to convert passive awareness reach into actual verified collection volume.

Publish a single open district-level e-waste dataset: HPSPCB/CPCB should consider releasing an annual, district-disaggregated generation/collection/compliance dataset so future assessments (including this one) can move from modelled estimates to verified figures.

13. Conclusion

Himachal Pradesh's score of 67.3/100 (GOOD) reflects a state with genuinely strong formal recycling capacity relative to its own generation, anchored in the Baddi-Barotiwala-Nalagarh industrial belt, but with a collection network, informal-sector formalisation programme and rural awareness infrastructure that have not yet caught up to that installed capacity. Closing the collection-rate gap (46.6% $\rightarrow$ closer to 100%), completing formalisation of the four known informal clusters, and converting passive awareness reach into active, verified take-back participation are the three highest-leverage moves available to push the state from “Good” into the “Excellent” band on this scorecard.

The complete underlying dataset, formulas and category-level detail behind every figure in this report are available in the companion workbook “Himachal Pradesh E-Waste Tracker.xlsx” and in the live dashboard linked on the cover page.